

Rui Zhao

Computer Science · Artificial Intelligence

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Personal Statement

I am a passionate and self-driven Computer Science undergraduate student specializing in AI. With research experience and learning ability far beyond my peers, I have a strong foundation in mathematics, artificial neural networks, reinforcement learning, large models and agent. While devoting myself to research, I also actively participate in projects and competitions in our school, which have honed my practical skills of coding and tool-using as well as cooperation abilities. I am interested in history, philosophy, literature and music so that I can think about the problems in the world from multiple perspectives and get inspiration from them. In a word, I am eager to learn and explore the unknown, and I am looking forward to expanding the frontiers of AI to make it more intelligent.

Research Interests

Reinforcement Learning, World Model and VLA, (Embodied Intelligence, LLM and Other Generative Models).

Education

2023 – **B.Eng. Department of Computer Science and Technology, Tsinghua University, Beijing, China**

Academic Performance

GPA: 3.9/4.0, **Rank:** 26/172

Research Projects

Compressing Generative World Models via Reinforcement Learning

- Tried to remove the noises and reserve task-relevant or reward-relevant information in world model by making the output more compressed
 - Designed a composite reward signal including compressibility, action prediction consistency (via an inverse action model trained on the feature of VLM), and value preservation to guide the generation process
 - Aimed to distill the fine-tuned generator into a significantly smaller model, enabling efficient world modeling by focusing only on essential visual information
- 2025 – now

RL for Reasoning

- Using the concepts in human cognitive psychology to design the reasoning process
- Constructing the cognitive graph / tree to represent the reasoning process and finding the better trajectory
- Using groups of samples with relative scores generated by DeepSeek R1 to train a smaller reward model
- Finetuning Qwen by RL to enhance general cognitive reasoning ability, not only on special tasks or datasets, and achieving better performance

Automated Construction of Knowledge Graph in Knowledge-intensive Domain

- Proposed an expandable framework of KG construction Tree-KG, including initial construction and iterative expansion
- 2024 – 2025 ○ Achieved sota performance compared with existing methods such as GraphRAG, especially when using textbooks
- Paper accepted at ACL 2025 main conference, Patent pending

Industrial Experience

- 2024 – 2025 **AI Teaching Assistant (Developer of Qingxiaozhi)**, *Tsinghua University*, Beijing, China
- Developed the core Meta Agent for the university's AI education system
- **Meta Agent Architecture:** Based on **OpenManus**, designed a high-level agent capable of formulating reusable, generalizable workflows for diverse teaching tasks (e.g., grading, course design) from single instructions
 - **Automated Tool & Workflow Construction:** Implemented a "Tool Maker" to synthesize Python scripts as new tools, and a "Workflow Maker" to organize operations into executable graphs, enabling the system to self-expand its capabilities
- Also worked on knowledge extraction: Extracting knowledge from textbooks and other materials to build a knowledge graph for the course

Honors and Scholarships

- Scholarship ShuPing Scholarship, 91-92, 93-94, 95-96
- Scholarship Academic Excellence Scholarship, 2024
- Scholarship Comprehensive Performance Scholarship, 2025
- Competition 3rd Place, Red Mountain Open-Source Large Model Innovation Competition, 2024
- Competition Second Prize, The 43th Tsinghua University "Challenge Cup" (Information Technology Track), 2025
- Competition 10th Place, "Rollman" - The 29th Intelligent Agent Competition (Ghost Track), 2025
- Honor Best Popularity Award, Student Innovation and Research Annual Conference, 2024

Skills

- Languages Python, C, C++, Rust, Typescript, Javascript, SystemVerilog
- Frameworks PyTorch, HuggingFace, verl, Weight & Biases, OpenCV
- Tools Git, Docker, Linux, LaTeX, VSCode

Publications

- [1] Songjie, N., et al. *Tree-KG: An Expandable KG Construction Framework for Knowledge-intensive Domains*. ACL 2025 main conference.

Languages

- Chinese Native
- English Fluent (CET-6: 611; TOEFL: Reading 30, Listening 30, Speaking N/A, Writing N/A)